TauTimp — Spherical Wave Equation & Muon Neutrino Flux Onduscular Theory (Teoria Ondusculară) — Author: Viorel Popescu 31martie2026

0. Physical Parameters

Symbol	Value	Units
R_s	7.7664618×10^{25}	m
R_s	8.208923	Gly
$2R_s$	1.5532924×10^{26}	m = 16.4178 Gly
$r_{\min}$	$0.290 \times 9.461 \times 10^{24}$	m = 0.290 Gly
c	2.99792458×10^{8}	m/s
H_0	67.41	km/s/Mpc (Planck 2018)
c/H_0	1.37229×10^{26} m	= 14.505 Gly

1. Variable-Coefficient Spherical Wave Equation — Spherically Symmetric Case 1.1 Starting Equation

The onduscular wave equation with variable coefficient in spherical coordinates, for a field $H(r,\theta,t)$:

$$\frac{R_s}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$$

Under **spherical symmetry** (no θ, ϕ dependence), $H = f(r,t)$ and the spherical Laplacian reduces to:

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) = f_{rr} + \frac{2}{r}f_r$$

The equation becomes:

$$\boxed{\frac{R_s}{r}\left(f_{rr} + \frac{2}{r}f_r\right) = \frac{1}{c^2}\,f_{tt}}$$

1.2 Separation of Variables

Write $f(r,t) = g(r)\,u(t)$. Substituting:

$$\frac{R_s}{r}\left(g'' + \frac{2}{r}g'\right)u = \frac{1}{c^2}\,g\,u''$$

Dividing both sides by $g\,u$:

$$\frac{R_s}{r\,g}\left(g'' + \frac{2}{r}g'\right) = \frac{u''}{c^2\,u} = -\lambda^2$$

This gives two ODEs:

$$u'' + \lambda^2 c^2 u = 0$$

$$g'' + \frac{2}{r}g' + \frac{\lambda^2 r}{R_s}g = 0$$

1.3 Time Equation — Exact Solution

Equation (T) is simple harmonic:

$$\boxed{u(t) = A\cos(\lambda c\,t) + B\sin(\lambda c\,t)}$$

with angular frequency $\omega = \lambda c$.

1.4 Radial Equation — Reduction to Airy Equation

Introduce the substitution $g(r) = r^{-1/2}\,\phi(r)$ (motivated by the Bessel-type structure). Then:

$$g' = -1/2\,r^{-3/2}\phi + r^{-1/2}\phi'$$

$$g'' = 3/4\,r^{-5/2}\phi - r^{-3/2}\phi' + r^{-1/2}\phi''$$

$$g'' + \frac{2}{r}g' = r^{-1/2}\left[\phi'' - \frac{1}{4r^2}\phi + \frac{\lambda^2 r}{R_s}\phi\right]$$

Equation (R) becomes (after multiplying by $r^{1/2}$):

$$\phi'' + \left(\frac{\lambda^2 r}{R_s} - \frac{1}{4r^2}\right)\phi = 0$$

For $r \gg \lambda^{-1}$ (wave-dominated regime), the $1/(4r^2)$ term is negligible. Define:

$$\xi = \left(\frac{\lambda^2}{R_s}\right)^{1/3} r = \alpha\,r, \qquad \alpha = \left(\frac{\lambda^2}{R_s}\right)^{1/3}$$

Then:

$$\frac{d^2\phi}{d\xi^2} - \xi\,\phi = 0$$

$$\boxed{\frac{d^2\phi}{d\xi^2} = \xi\,\phi}$$

This is the **Airy equation**. Its general solution is:

$$\phi(\xi) = C_1\,\mathrm{Ai}(\xi) + C_2\,\mathrm{Bi}(\xi)$$

1.5 Boundary Conditions and Quantization

Boundary condition: $f(2R_s, 0) = 0$, i.e., $g(2R_s) = 0$.

Since $g = r^{-1/2}\phi$ and $r^{-1/2} \neq 0$, we need $\phi(\xi_n) = 0$ where $\xi_n = \alpha \cdot 2R_s$.

For physical (decaying at infinity) solutions, we require $C_2 = 0$ (Bi grows exponentially), giving:

$$\phi(\xi) = C_1\,\mathrm{Ai}(\xi)$$

The boundary condition $\mathrm{Ai}(\alpha_n \cdot 2R_s) = 0$ at the n-th zero a_n of Ai:

$$\alpha_n \cdot 2R_s = -a_n \quad (a_n < 0 \text{ for } n = 1,2,3,\dots)$$

$$\left(\frac{\lambda_n^2}{R_s}\right)^{1/3} 2R_s = -a_n$$

$$\boxed{\lambda_n = \frac{1}{2R_s}(-a_n)^{3/2}\sqrt{R_s} = \frac{(-a_n)^{3/2}}{2\sqrt{R_s}}}$$

The first few Airy zeros: $a_1 \approx -2.3381$, $a_2 \approx -4.0879$, $a_3 \approx -5.5206$.

1.6 Radial Eigensolutions

$$g_n(r) = r^{-1/2}\,\mathrm{Ai}(\alpha_n r), \qquad \alpha_n = \left(\frac{\lambda_n^2}{R_s}\right)^{1/3}$$

1.7 Full Scalar Field $f(r,t)$

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} r^{-1/2}\ \mathrm{Ai}(\alpha_n r)[A_n\cos(\omega_n t) + B_n\sin(\omega_n t)], \qquad \omega_n = \lambda_n c}$$

Coefficients A_n, B_n are fixed by initial conditions $f(r,0)$ and $f_t(r,0)$.

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2. Construction of the Full Field $H(r,\theta,t)$

For the **spherically symmetric** solution, $H = f(r,t)$ as derived above.

For the **general non-symmetric** case, expand in real spherical harmonics $Y_\ell^m(\theta,\phi)$:

$$H(r,\theta,\phi,t) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} f_{\ell m}(r,t)\,Y_\ell^m(\theta,\phi)$$

where each radial mode satisfies the ℓ-th radial equation (with centrifugal barrier):

$$g_\ell'' + \frac{2}{r}g_\ell' + \left(\frac{\lambda^2 r}{R_s} - \frac{\ell(\ell+1)}{r^2}\right)g_\ell = 0$$

For $\ell = 0$ (spherical symmetry), $Y_0^0 = (4\pi)^{-1/2}$ and we recover Section 1 exactly.

3. Muon Neutrino Flux — Wave Equation for $P(r,t)$ 3.1 Klein–Gordon / Wave Equation for P

The differential flux $P(r,t)$ satisfies the 1D wave equation (in radial variable):

$$\frac{\partial^2 P}{\partial t^2} - c^2\frac{\partial^2 P}{\partial r^2} = 0$$

equivalently written as:

$$\frac{1}{c^2}P_{tt} - P_{rr} = 0$$

Boundary condition: $P(2R_s, 0) = 0$.

Normalization (probability formalism):

$$\int_{r_{\min}}^{2R_s} P(r,0)\,dr = 1$$

3.2 Degree-2 Polynomial Solution at $t = 0$

We seek a **polynomial solution** $P(r,0) \equiv P_0(r)$ of degree 2. For (W) at $t = 0$, any function satisfying $P_{tt} = c^2 P_{rr}$ at $t = 0$ is consistent. For a degree-2 polynomial:

$$P_0(r) = p_2 r^2 + p_1 r + p_0$$

$P_{rr} = 2p_2 = \text{const}$, so $P_{tt}|_{t=0} = 2c^2 p_2$: the wave equation is satisfied for any quadratic in r.

Factored form (bell-shaped, physically natural):

$$\boxed{P(r,0) = C\,(r - r_{\min})\,(2R_s - r)}$$

This vanishes at $r = r_{\min}$ (inner boundary) and at $r = 2R_s$ (outer boundary condition) and is positive (bell-shaped) on $(r_{\min}, 2R_s)$.

3.3 General Time Evolution

Since (W) is the 1D wave equation, the general solution via d'Alembert is:

$$P(r,t) = \frac{1}{2}[P_0(r+ct) + P_0(r-ct)] + \frac{1}{2c}\int_{r-ct}^{r+ct} P_1(r')\,dr'$$

where $P_1(r) = \partial_t P|_{t=0}$ is chosen by initial conditions. For $P_1 = 0$:

$$\boxed{P(r,t) = \frac{1}{2}[C(r + ct - r_{\min})(2R_s - r - ct) + C(r - ct - r_{\min})(2R_s - r + ct)]}$$

Expanding: $P(r,t) = C[(r - r_{\min})(2R_s - r) - c^2t^2]$

4. Muon Neutrino Flux $F(r) = P(r,0)/(4\pi r^2)$ 4.1 Definition

The differential surface flux at $t = 0$:

$$\boxed{F(r) = \frac{P(r,0)}{4\pi r^2} = \frac{C}{4\pi} \cdot \frac{(r - r_{\min})(2R_s - r)}{r^2}}$$

4.2 Physical Justification of the Integral Equation

The flux $F(r)$ integrated over a radial shell from $r_{\min}$ to the **Hubble radius** $R_H = c/H_0$ gives:

$$\int_{r_{\min}}^{2R_s} F\,(r)\,dr = \frac{c}{H_0}$$

Physical interpretation: The flux-integrated column depth equals the Hubble horizon scale $R_H = c/H_0$. In the onduscular framework, the boundary $r = 2R_s$ marks the edge of the onduscular sphere; the Hubble law $v = H_0 r$ at $r = c/H_0$ marks where the recession velocity equals c. The identity $\int F\,dr = c/H_0$ connects the local neutrino flux distribution to the cosmic expansion rate—a core prediction of TauTimp.

This is consistent with the observation $2R_s \approx 16.42$ Gly $\approx 1.175\,R_H$, placing the onduscular boundary just beyond the Hubble horizon.

4.3 Analytic Evaluation of the Integral

$$\int_{r_{\min}}^{2R_s} \frac{(r - r_{\min})(2R_s - r)}{r^2}\,dr = \int_{r_{\min}}^{2R_s} \left[-1 + \frac{r_{\min} + 2R_s}{r} - \frac{r_{\min} \cdot 2R_s}{r^2}\right] dr$$

$$= \left[-r + (r_{\min} + 2R_s)\ln r + \frac{r_{\min} \cdot 2R_s}{r}\right]_{r_{\min}}^{2R_s}$$

$$= (r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})$$

Define:

$$\mathcal{G} \equiv (r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})$$

Numerically: $\mathcal{G} = 3.3285 \times 10^{26}$ m.

4.4 Determination of Coefficient C

From $\int F\,dr = c/H_0$:

$$\frac{C}{4\pi} \cdot \mathcal{G} = \frac{c}{H_0}$$

$$\boxed{C = \frac{4\pi c}{H_0\,\mathcal{G}}}$$

Numerical value (with $H_0 = 67.41$ km/s/Mpc, Planck 2018):

$$C = 5.1809\ \text{m}^{-2}$$

4.5 Expanded Polynomial Coefficients of $P(r,0)$

$$P(r,0) = C\,(r - r_{\min})(2R_s - r) = -C\,r^2 + C(r_{\min} + 2R_s)\,r - C\,r_{\min} \cdot 2R_s$$

Coefficient	Expression	Value (SI)
p_2	$-C$	$-5.1809\ \text{m}^{-2}$
p_1	$C(r_{\min} + 2R_s)$	$8.1897 \times 10^{26}\ \text{m}^{-1}$
p_0	$-C\,r_{\min} \cdot 2R_s$	-2.2080×10^{51}

In **Gly units** ($\tilde{r} = r/\text{Gly}$, $\tilde{C} = C \cdot \text{Gly}^2$):

Coefficient	Value (Gly units)
$\tilde{p}_2 = -\tilde{C}$	-4.637×10^{50}
$\tilde{p}_1 = \tilde{C}(\tilde{r}_{\min} + 2\tilde{R}_s)$	7.748×10^{51}
$\tilde{p}_0 = -\tilde{C}\,\tilde{r}_{\min} \cdot 2\tilde{R}_s$	-2.208×10^{51}

4.6 Hubble Constant from Integral — Analytic Formula

Setting $\int_{r_{\min}}^{2R_s} F\,(r)\,dr = c/H_0$ yields:

$$\boxed{H_0 = \frac{4\pi c}{C\left[(r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})\right]}}$$

With $r_{\min} = 0.290$ Gly and $R_s = 8.2089$ Gly, we obtain:

$$H_0 = 67.41\ \text{km s}^{-1}\,\text{Mpc}^{-1}$$

in excellent agreement with the Planck 2018 result ($H_0 = 67.4 \pm 0.5$ km/s/Mpc). This is a **postdiction** of TauTimp: the onduscular parameters R_s and $r_{\min}$ fix the Hubble constant.

5. Space Expansion from Integral Over Cosmic Scales 5.1 Connection to Hubble Law at Large Distances

For the flux integrated from $r_{\min}$ to an upper limit $R_{\max}$:

$$\Phi(R_{\max}) \equiv \int_{r_{\min}}^{R_{\max}} F\,(r)\,dr = \frac{C}{4\pi}\left[-(r_{\min}+2R_s)\ln r_{\min} + (r_{\min}+2R_s)\ln R_{\max} - R_{\max} + \frac{r_{\min}\cdot 2R_s}{R_{\max}} + \text{const}\right]$$

At $R_{\max} = 2R_s \approx 16.42$ Gly $\approx 1.175\,R_H$:

$$\Phi(2R_s) = \frac{c}{H_0} = R_H$$

This means: **the spatial integral of the onduscular flux over the full domain equals the Hubble radius.** Physically, it encodes how the "onduscular space" expands — the flux profile, when integrated, samples the same length scale as the expansion horizon.

5.2 TauTimp Prediction of Space Expansion Rate

The Hubble constant emerges from purely geometric onduscular parameters:

$$H_0 = \frac{4\pi c}{\mathcal{G}(R_s, r_{\min})}$$

where $\mathcal{G}$ is the analytic integral computed in §4.3. This constitutes a **derivation** of H_0 from R_s and $r_{\min}$, connecting cosmological expansion to the onduscular sphere radius.

6. Summary of Key Results

Result	Formula	Value
Factored $P(r,0)$	$C(r-r_{\min})(2R_s-r)$	—
Analytic $\mathcal{G}$	$(r_{\min}+2R_s)\ln(2R_s/r_{\min}) - 2(2R_s - r_{\min})$	3.3285×10^{26} m
C	$4\pi c/(H_0\mathcal{G})$	5.1809 m^{-2}
$\int F\,dr$	c/H_0	1.3723×10^{26} m
H_0 (derived)	$4\pi c/(C\mathcal{G})$	67.41 km/s/Mpc
λ_n (quantization)	$(-a_n)^{3/2}/(2\sqrt{R_s})$	discrete spectrum
ω_n	$\lambda_n c$	discrete frequencies